

Cycle 17 Autonomous Report

Date: 2026-05-16

Repository: qc_exciton_LCC

Task selected

Add a reusable autonomous-cycle PDF report generator so the inbox requirement can be satisfied without copying a cycle-specific script each time.

Major changes

Added scripts/autonomous_cycle_report.py with matplotlib mathtext equations and argument-driven report sections.

Added a smoke test that writes a non-empty PDF in a temporary directory.

Files changed

scripts/autonomous_cycle_report.py

tests/test_autonomous_cycle_report.py

docs/cycle_reports/cycle17_report.pdf

Rendered Equations

This page is a rendering check for the inbox requirement that equations appear as LaTeX-style math.

$$T(M, N, K, b) = K[2(\log_2 N + 1)b + I_1(M, N)]$$

$$Q(M, N, b) = \lceil \log_2 M \rceil + 1 + \log_2 N + b + W(M, N, b)$$

$$T(M)/M \rightarrow 0 \quad \text{for sublinear QROAMClean amortization}$$

Goals already achieved

Cycle 16 already made the synthesis estimator tables regenerable from the Bloq.

Cycles 5-15 already closed the synthesis analytic-vs-Bloq Toffoli/qubit report path over the tabulated grid.

Verification

Tests and checks

PYTHONPATH=src python tests/test_autonomous_cycle_report.py -> 1/1 passed

Achieved goal

Achieved goal: Improved report-generation infrastructure for final Toffoli complexity/qubit count/scaling work by making per-cycle PDF delivery reusable and math-rendered.

Next recommended task

Resume deriving $I_1(n_blocks, n_rows)$ in closed form from QROAMClean optimizer behavior, using scripts/regenerate_synthesis_tables.py as the regression oracle.